GLOBAL INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GR25

PROBLEM SOLVING THROUGH C PROGRAMMING LAB MANUAL

Lab Syllabus

Simple numeric problems:

1. Write a program for finding the max and min from the three numbers.
2. Write the program for the simple, compound interest.
3. Write a program that prints a multiplication table for a given number and the number of rows in the table. For example, for a number 5 and rows = 3, the output should be:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
4. Write a program that shows the binary equivalent of a given positive number between 0 to 255.

Expression Evaluation:

5. Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators +,-,*, /, % and use Switch Statement).
6.Write a program that finds if a given number is a prime number.
7.Write a C program to find the sum of individual digits of a positive integer and test given number is palindrome.
8.A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.

Arrays, Pointers and Functions:

9. Write a C program to find the minimum, maximum and average in an array of integers.
10 Write a C program that uses functions to perform the following:
I. Addition of Two Matrices
10 II. Multiplication of Two Matrices
11. display the values using the array.
12. Write a program for display values reverse order from an array using a pointer.

Files:

13. Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.
14. Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

Strings:

15. Write a C program that uses functions to perform the following operations:
I. To insert a sub-string into a given main string from a given position.
15 II. To delete n Characters from a given position in a given string
16. Write a C program to determine if the given string is a palindrome or not (Spelled same in both directions with or without a meaning like madam, civic, noon, abcba, etc.)
17. Write a C program that displays the position of a character ch in the string S or – 1 if S doesn't contain ch.
18. Write a C program to count the lines, words and characters in a given text.

Sorting and Searching:

19. Write a C program that uses non-recursive function to search for a Key value in a given list of integers using linear search method.
20. Write a C program that uses non-recursive function to search for a Key value in a given sorted list of integers using binary search method.
21. Write a C program that implements the Bubble sort method to sort a given list of integers in ascending order.
22. Write a C program that sorts the given array of integers using selection sort in descending order
23. Write a C program that sorts the given array of integers using insertion sort in ascending order
24. Write a C program that sorts a given array of names.

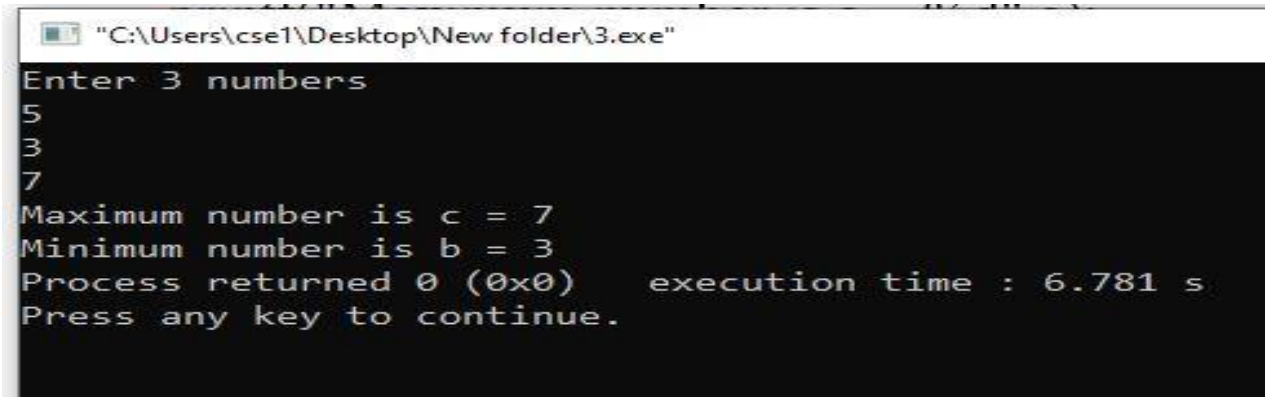
Simple Numeric Problems:

1. Write a program for finding the max and min from the three numbers.

Program:

```
#include<stdio.h>
int main()
{
int a,b,c;
printf("Enter 3 numbers");
scanf("%d%d%d",&a,&b,&c);
if(a>b && a>c)
printf("Maximum number is a = %d",a);
else if(b>a && b>c)
printf("Maximum number is b = %d",b);
else
printf("Maximum number is c = %d",c);
printf("\n");
if(a<b && a<c)
printf("Minimum number is a = %d",a);
else if(b<a && b<c)
printf("Minimum number is b = %d",b);
else
return 0;
}
```

Output:



2. Write the program for the simple, compound interest.

Program:

```
#include <stdio.h>
#include <math.h>
int main() {
   float P, rate, time, SI, CI;
   // Input P amount, rate of interest and time
   printf("Enter Principal amount ,Rate of interest ,Time: ");
   scanf("%f%f%f", &P,&rate,&time);
   // Calculate simple interest
   SI = (P * rate * time) / 100;
   // Calculate compound interest
   CI = P * pow(1 + (rate / 100), time) - P;
   // Display results
   printf("SI= %f\n", SI);
   printf("CI= %f\n", CI);
   return 0;
}
```

Output:

```
Enter Principal amount ,Rate of interest ,Time: 5000
10
24
SI= 12000.000000
CI= 44248.687500

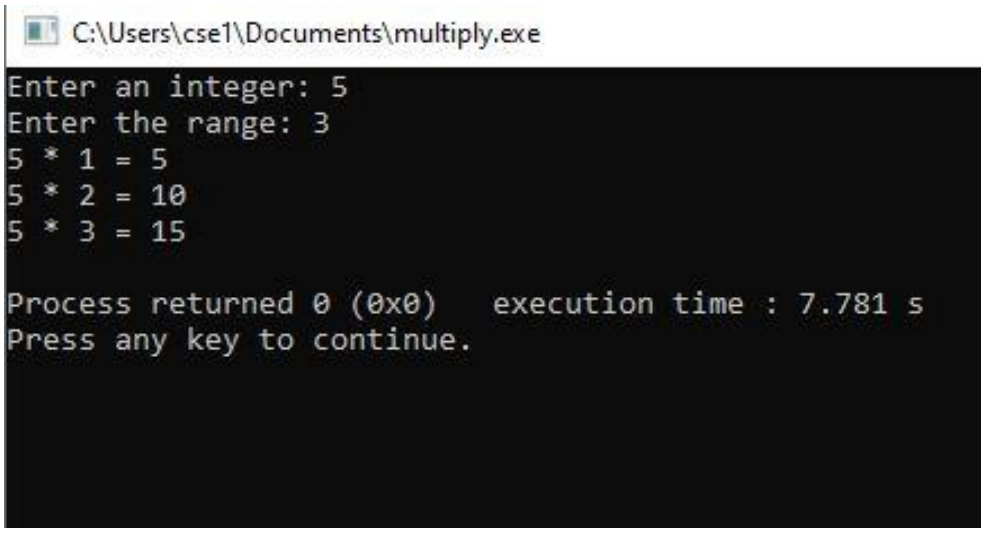

...Program finished with exit code 0
Press ENTER to exit console.
```

3. Write a program that prints a multiplication table for a given number and the number of rows inthe table. For example, for a number 5 and rows = 3, the output should be:

Program:

```
#include <stdio.h>
int main()
{
int n, i, range;
printf("Enter an integer: ");
 scanf("%d",&n);
printf("Enter the range: ");
scanf("%d", &range);
for(i=1; i<=range;i++)
{
printf("%d * %d = %d \n",n,i, n*i);
}
return 0;
}
```

Output:

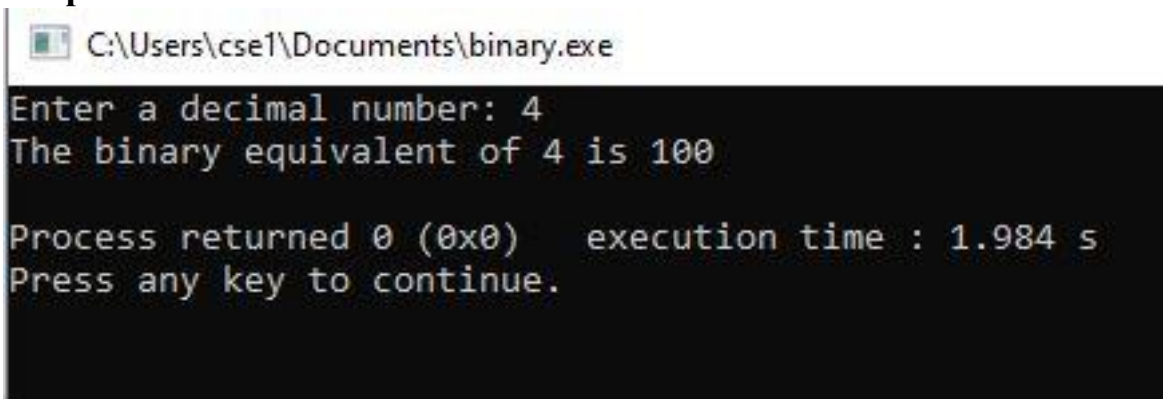


4. Write a program that shows the binary equivalent of a given positive number between 0 to 255.

Program:

```
#include <stdio.h>
int binary_conversion(int);
int main()
{
int num, bin;
printf("Enter a decimal number: ");
scanf("%d", &num);
bin = binary_conversion(num);
printf("The binary equivalent of %d is %d\n", num, bin);
}
int binary_conversion(int num)
{
if (num == 0)
{
return 0;
}
else
{
return (num % 2) + 10 * binary_conversion(num / 2);
return 0;
}
}
```

Output:



Expression Evaluation:

5. Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators +,-,*, /, % and use Switch Statement)

Program:

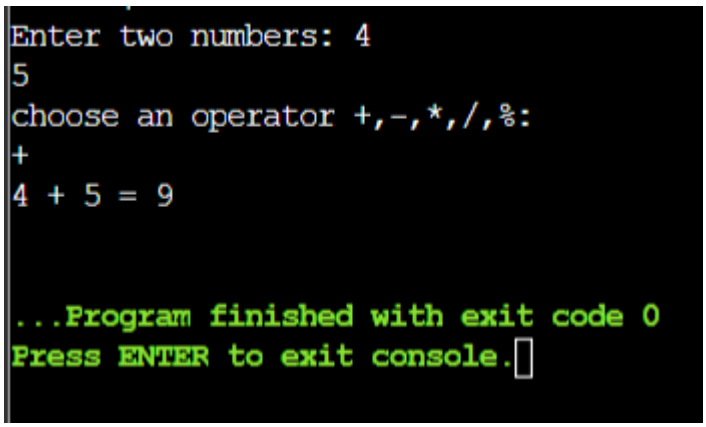
```
#include <stdio.h>
int main()
{
   int num1, num2, result;
   char operator;
   // Take input from the user
   printf("Enter two numbers: ");
   scanf("%d%d", &num1,&num2);
   printf("choose an operator +,-,*,/,%: ");
   scanf(" %c", &operator);  // Note the space before %c to consume any whitespace

   // Perform the operation based on the operator
   switch (operator) {
      case '+':
         result = num1 + num2;
         printf("%d + %d = %d\n", num1, num2, result);
         break;
      case '-':
         result = num1 - num2;
         printf("%d - %d = %d\n", num1, num2, result);
         break;
      case '*':
         result = num1 * num2;
         printf("%d * %d = %d\n", num1, num2, result);
         break;
      case '/':
         if (num2 != 0) {
            result = num1 / num2;
            printf("%d / %d = %d\n", num1, num2, result);
         } else {
            printf("Error: Division by zero is not allowed.\n");
         }
         break;
      case '%':
         if (num2 != 0) {
            result = num1 % num2;
            printf("%d %% %d = %d\n", num1, num2, result);
         } else {
```

```
            printf("Error: Division by zero is not allowed for modulus.\n");
        }
        break;
    default:
        printf("Invalid operator\n");
        break;
  }

  return 0;
}
```

Output:



6. Write a program that finds if a given number is a prime number

Program:

```
#include <stdio.h>
int main() {
   int number,count=0;

   printf("Enter a number: ");
   scanf("%d", &number);
   for (int i = 2; i<= number/2; i++) {
      if (number % i == 0) {
         count++;   // Found a divisor, not prime
         break;
      }
   }
   if (count == 0) {
      printf("%d is a prime number.\n", number);
   } else {
      printf("%d is not a prime number.\n", number);
   }

   return 0;
}
```

Output:

```
Enter a number: 5
5 is a prime number.


...Program finished with exit code 0
Press ENTER to exit console.
```

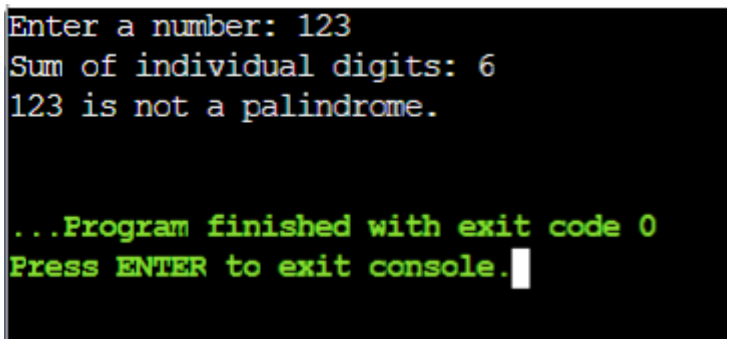
7. Write a C program to find the sum of individual digits of a positive integer and test given number is palindrome.

Program:

```
#include <stdio.h>
int main() {
   int num,sum = 0,reversed,temp,digit;
   printf("Enter a number: ");
   scanf("%d", &num);
   temp = num;
   while (temp > 0) {
      digit=temp%10;
      sum += digit;  // Add the last digit to sum
      reversed = reversed * 10 + digit;  // Build the reversed number
      temp /= 10;  // Remove the last digit
   }
   printf("Sum of individual digits: %d\n", sum);
   // Check if given number and reversed are the same
   if (num == reversed) {
       printf("%d is a palindrome.\n", num);
   } else {
      printf("%d is not a palindrome.\n", num);
   }

   return 0;
}
```

Output:

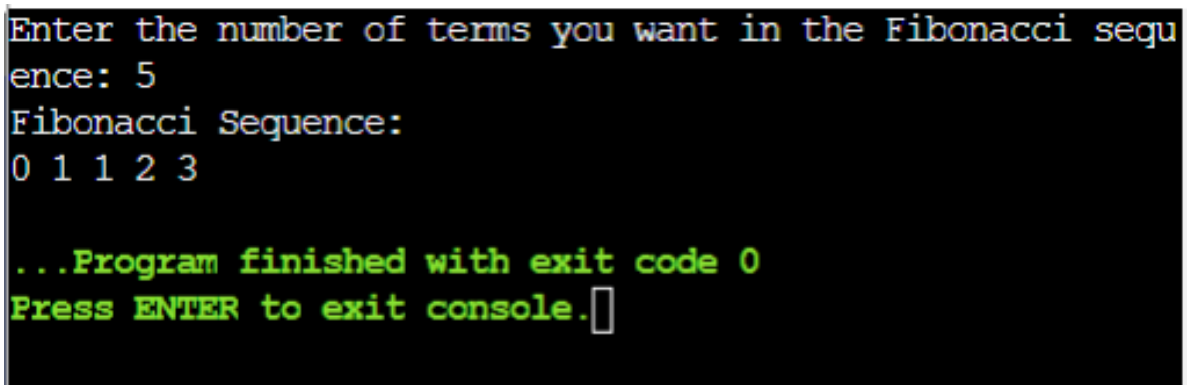


8. A Fibonacci sequence is defined as follows: the first and second terms in the sequence are 0 and 1. Subsequent terms are found by adding the preceding two terms in the sequence. Write a C program to generate the first n terms of the sequence.

Program:

```
#include <stdio.h>
int main() {
   int n, a = 0, b = 1, next,i;
   printf("Enter the number of terms you want in the Fibonacci sequence: ");
   scanf("%d", &n);
   printf("Fibonacci Sequence:\n");
   printf("%d %d ",a,b);
   for (i = 3; i <= n; i++) {
      next = a + b;
      printf("%d ", next);
      a = b;
      b = next;
   }
   return 0;
}
```

Output:



Arrays, Pointers and Functions:

9.Write a C program to find the minimum, maximum and average in an array of integers.

Program:

```
#include <stdio.h>
#include <math.h>
int main() {
    int n,arr[10],min,max,sum;
    float average;
    // Read the size of the array
    printf("Enter the number of elements in the array: ");
    scanf("%d", &n);
    // Read the elements of the array
    printf("Enter %d integers:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    // Initialize min, max and sum
    min = arr[0];
    max = arr[0];
    sum = arr[0]; // Using long long to avoid overflow
    // Iterate through the array to find min, max, and sum
    for (int i = 1; i < n; i++) {
        if (arr[i] < min) {
            min = arr[i];
        }
        if (arr[i] > max) {
            max = arr[i];
        }
        sum += arr[i];
    }
    // Calculate the average
    average = (float)sum / n;
    // Print the results
    printf("Minimum: %d\n", min);
    printf("Maximum: %d\n", max);
    printf("Average: %.2f\n", average);
    return 0;
}
```

Output:

```
Enter the number of elements in the array: 5
Enter 5 integers:
12
45
32
10
5
Minimum: 5
Maximum: 45
Average: 20.80


...Program finished with exit code 0
Press ENTER to exit console.
```

10. Write a C program that uses functions to perform the following:

I. Addition of Two Matrices

Program

```
#include <stdio.h>
// Function to perform matrix addition
void addMatrices(int A[10][10], int B[10][10], int result[10][10], int rows, int cols) {
    int i,j;
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            result[i][j] = A[i][j] + B[i][j];
        }
    }
}
// Function to input matrix elements
void inputMatrix(int matrix[10][10], int rows, int cols) {
    printf("Enter elements of the matrix (%dx%d):\n", rows, cols);
    int i,j;
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }
}
// Function to display a matrix
void displayMatrix(int matrix[10][10], int rows, int cols) {
    int i,j;
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            printf("%2d ", matrix[i][j]);
        }
        printf("\n");
    }
}
int main() {
    int A[10][10], B[10][10], result[10][10];
    int rows, cols;
    // Input matrix dimensions
    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &cols);
    // Input matrices A and B
    printf("Input matrix A:\n");
    inputMatrix(A, rows, cols);
    printf("Input matrix B:\n");
    inputMatrix(B, rows, cols);
```

```
    // Add matrices A and B
    addMatrices(A, B, result, rows, cols);
    //Display the results
    printf("\nMatrix A:\n");
    displayMatrix(A, rows, cols);
    printf("\nMatrix B:\n");
    displayMatrix(B, rows, cols);
    printf("\nResultant Matrix (A + B):\n");
    displayMatrix(result, rows, cols);
    return 0;
}
```

Output:

```
Enter the number of rows and columns:
2
2
Input matrix A:
Enter elements of the matrix (2x2):
2 3
4 5
Input matrix B:
Enter elements of the matrix (2x2):
7 5
2 1

Matrix A:
 2  3
 4  5

Matrix B:
 7  5
 2  1

Resultant Matrix (A + B):
 9  8
 6  6
```

10.II. Multiplication of Two Matrices.

Program:

```
#include <stdio.h>
// Function to perform matrix multiplication
void multiplyMatrices(int A[10][10], int B[10][10], int result[10][10], int rowA, int colA, int
rowB, int colB) {
    int i,j,k;
    for (i = 0; i < rowA; i++) {
        for (j = 0; j < colB; j++) {
            result[i][j] = 0;           // Initializing the result matrix with 0
            for (k = 0; k < colA; k++) {
                result[i][j] += A[i][k] * B[k][j];
            }
        }
    }
}

// Function to input matrix elements
void inputMatrix(int matrix[10][10], int rows, int cols) {
    int i,j;
    printf("Enter elements of the matrix (%dx%d):\n", rows, cols);
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }
}

// Function to display a matrix
void displayMatrix(int matrix[10][10], int rows, int cols) {
    int i,j;
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            printf("%d ", matrix[i][j]);
        }
        printf("\n");
    }
}

int main() {
    int A[10][10], B[10][10], result[10][10];
    int rowA, colA, rowB, colB;

    // Input matrix dimensions
    printf("Enter the number of rows and columns for Matrix A: ");
```

```
    scanf("%d %d", &rowA, &colA);

    printf("Enter the number of rows and columns for Matrix B: ");
    scanf("%d %d", &rowB, &colB);

    // Check if multiplication is possible (columns of A must be equal to rows of B)
    if (colA != rowB) {
        printf("Matrix multiplication is not possible. The number of columns of A must be equal
to the number of rows of B.\n");
        return 1; // Exit the program if multiplication is not possible
    }

    // Input matrices A and B
    printf("Input matrix A:\n");
    inputMatrix(A, rowA, colA);

    printf("Input matrix B:\n");
    inputMatrix(B, rowB, colB);

    // Perform matrix multiplication
    multiplyMatrices(A, B, result, rowA, colA, rowB, colB);

    // Display the result
    printf("\nMatrix A:\n");
    displayMatrix(A, rowA, colA);

    printf("\nMatrix B:\n");
    displayMatrix(B, rowB, colB);

    printf("\nResultant Matrix (A * B):\n");
    displayMatrix(result, rowA, colB);

    return 0;
}
```

Output:

```
Enter the number of rows and columns for Matrix A:
2
2
Enter the number of rows and columns for Matrix B: 2
2
Input matrix A:
Enter elements of the matrix (2x2):
2 1
3 4
Input matrix B:
Enter elements of the matrix (2x2):
7 8
1 2

Matrix A:
2 1
3 4

Matrix B:
7 8
1 2

Resultant Matrix (A * B):
15 18
25 32
```

11. Write a program for reading elements using a pointer into an array and display the values using the array.

Program:

```
#include <stdio.h>
int main() {
   int n, sum = 0, arr[10];
   // Input the number of elements
   printf("Enter the number of elements: ");
   scanf("%d", &n);
   int *ptr = arr;
    printf("Enter the elements:\n");
   for (int i = 0; i < n; i++) {
      // Read element using the pointer
      scanf("%d", ptr + i);  // Pointer arithmetic to read the value
   }

   // Display the values using the array
   printf("The elements are:\n");
   for (int i = 0; i < n; i++) {
      printf("%d ", arr[i]);  // Display elements using the array
   }
   return 0;
}
```

Output:

```
Enter the number of elements: 5
Enter the elements:
10
45
1
21
5
The elements are:
10 45 1 21 5
```

12. Write a program for display values reverse order from an array using a pointer.

Program:

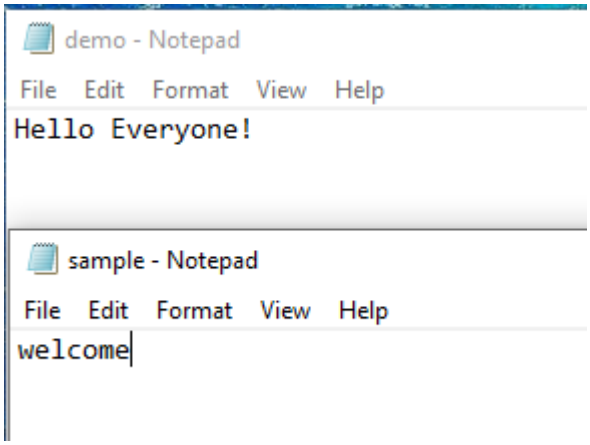
```
#include <stdio.h>
int main() {
   int n, sum = 0, arr[10];
   // Input the number of elements
   printf("Enter the number of elements: ");
   scanf("%d", &n);
   // Input the array elements
   printf("Enter the elements:\n");
   for (int i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }
   int *ptr = arr;
   // Display the array elements in reverse order
   printf("The elements in reverse order are:\n");
   for (int i = n-1; i >= 0; i--) {
     printf("%2d", *(ptr+i));
   }
   return 0;
}
```

Output:

```
Enter the number of elements: 4
Enter the elements:
2
4
6
1
The elements in reverse order are:
 1 6 4 2
```

Files:

13. Write a C program which copies one file to another, replacing all lowercase characters with their uppercase equivalents.

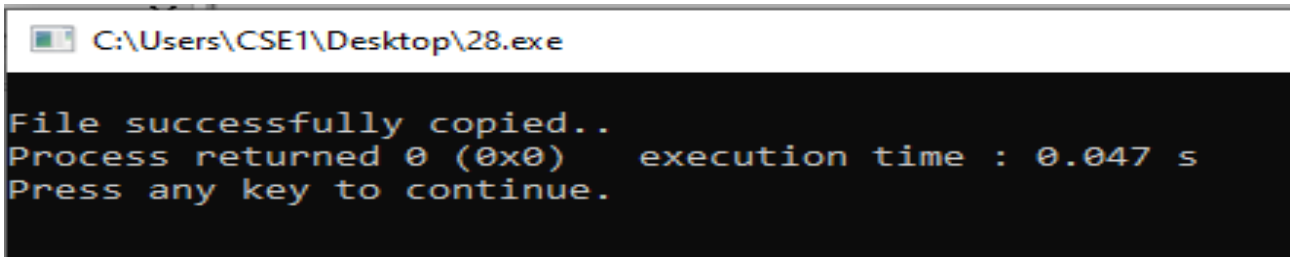


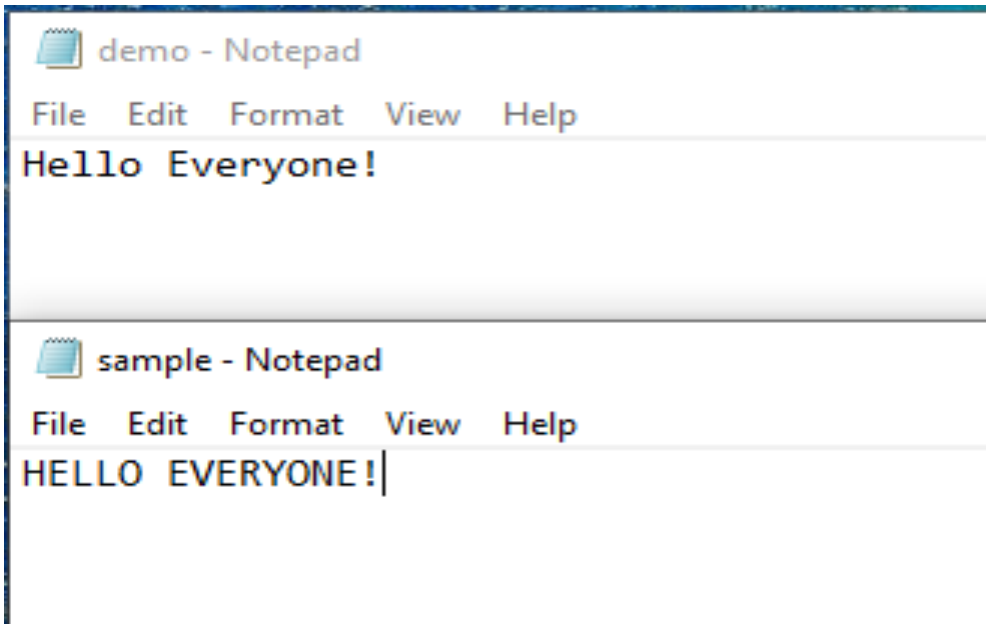
Program:

```
#include<stdio.h>
#include<stdlib.h>
#include<ctype.h>
int main()
{
    FILE *fp1, *fp2;
    char ch;
    fp1 = fopen("demo.txt", "r");
    if (fp1 == NULL)
    {
        puts("File does not exist..");
        exit(1);
    }
    fp2 = fopen("sample.txt", "w");
    if (fp2 == NULL)
    {
        puts("File does not exist..");
        fclose(fp1);
        exit(1);
    }
    while((ch=fgetc(fp1))!=EOF)
    {
        ch = toupper(ch);
        fputc(ch,fp2);
    }
    printf("\nFile successfully copied..");
    return 0;
```

}

Output:


C:\Users\CSE1\Desktop\28.exe
File successfully copied..
Process returned 0 (0x0) execution time : 0.047 s
Press any key to continue.


demo - Notepad
File Edit Format View Help
Hello Everyone!
sample - Notepad
File Edit Format View Help
HELLO EVERYONE!

14. Write a C program to merge two files into a third file (i.e., the contents of the first file followed by those of the second are put in the third file).

Program:

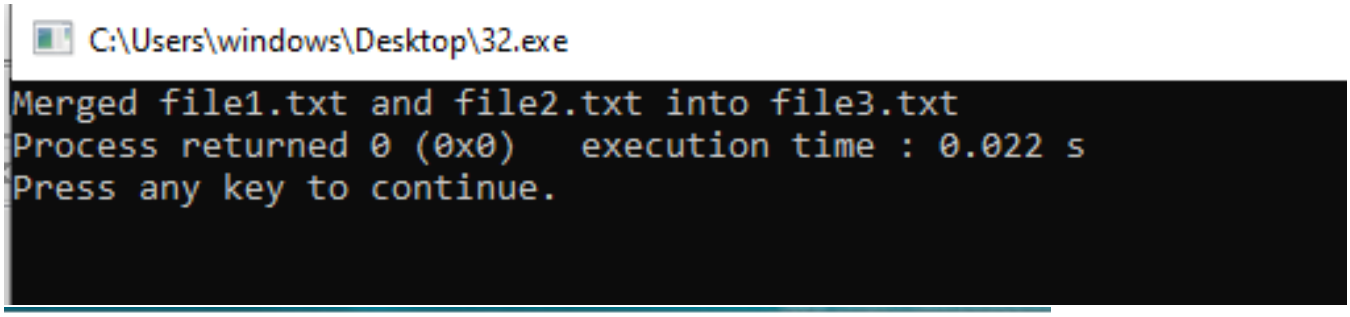
```
#include <stdio.h>
#include <stdlib.h>
int main()
{
  // Open two files to be merged
  FILE *fp1 = fopen("file1.txt", "r");
  FILE *fp2 = fopen("file2.txt", "r");

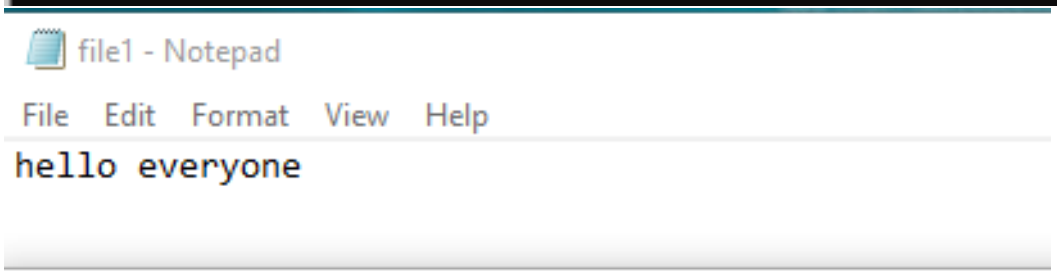
  // Open file to store the result
  FILE *fp3 = fopen("file3.txt", "w");
  char c;

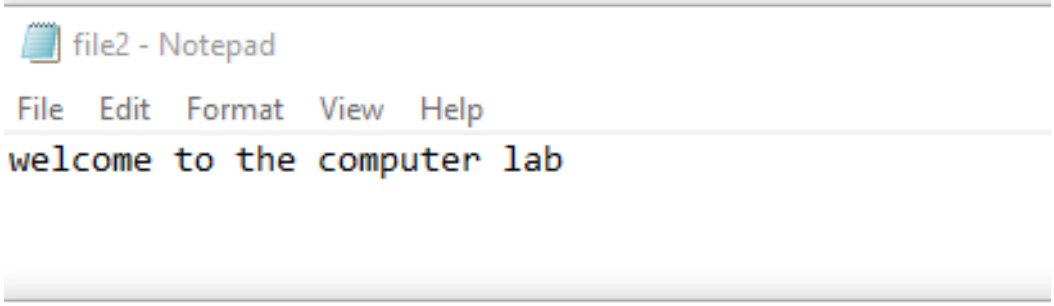
  if (fp1 == NULL || fp2 == NULL || fp3 == NULL)
  {
      puts("Could not open files");
      exit(0);
  }
  // Copy contents of first file to file3.txt
  while ((c = fgetc(fp1)) != EOF)
    fputc(c, fp3);

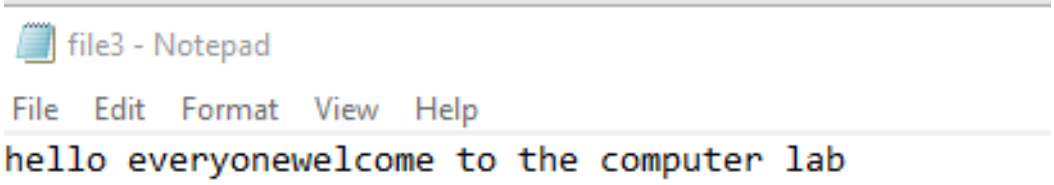
  // Copy contents of second file to file3.txt
  while ((c = fgetc(fp2)) != EOF)
    fputc(c, fp3);
  printf("Merged file1.txt and file2.txt into file3.txt");
  fclose(fp1);
  fclose(fp2);
  fclose(fp3);
  return 0;
}
```

Output:


C:\Users\windows\Desktop\32.exe
Merged file1.txt and file2.txt into file3.txt
Process returned 0 (0x0)   execution time : 0.022 s
Press any key to continue.


file1 - Notepad
File Edit Format View Help
hello everyone


file2 - Notepad
File Edit Format View Help
welcome to the computer lab


file3 - Notepad
File Edit Format View Help
hello everyonewelcome to the computer lab

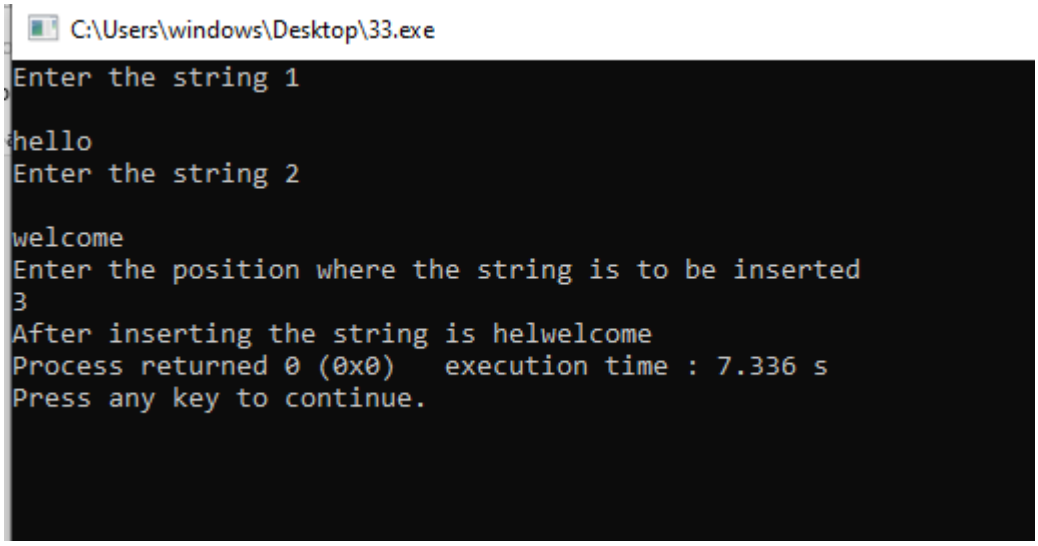
Strings

15. Write a C program that uses functions to perform the following operations
I. To insert a sub-string into a given main string from a given position.

Program:

```
#include<stdio.h>
#include<string.h>
int main()
{
  char str1[20], str2[20];
  int l1, l2, n, i;
  puts("Enter the string 1\n");
  gets(str1);
  l1 = strlen(str1);
  puts("Enter the string 2\n");
  gets(str2);
  l2 = strlen(str2);
  printf("Enter the position where the string is to be inserted\n");
  scanf("%d", &n);
  for(i = n; i < 20; i++)
 {
    str1[i + 20] = str1[i];
  }
  for(i = 0; i < 20; i++)
  {
    str1[n + i] = str2[i];
  }
    str2[20+ 1] = '\0';
  printf("After inserting the string is %s", str1);
  return 0;
}
```

Output:

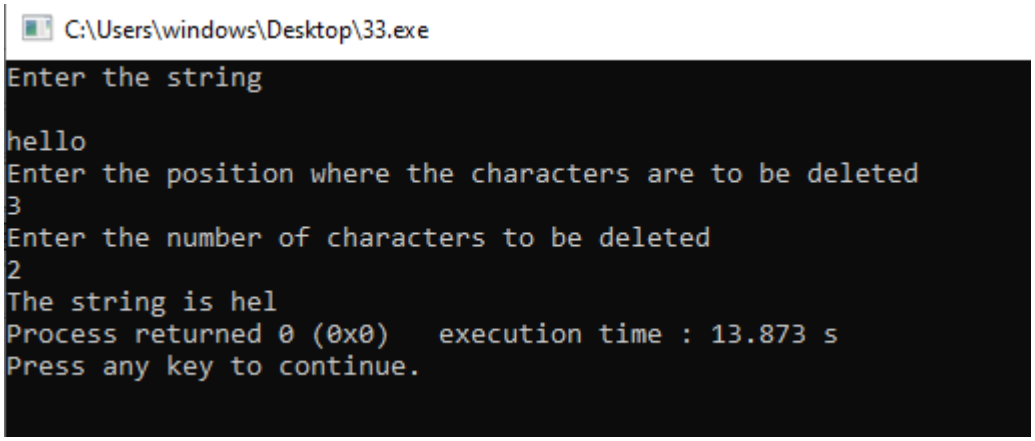


15 II. To delete n Characters from a given position in a given string.

Program:

```
#include<stdio.h>
#include<string.h>
int main()
{
  char str[20];
  int i, n, l, pos;
  puts("Enter the string\n");
  gets(str);
  printf("Enter the position where the characters are to be deleted\n");
  scanf("%d", &pos);
  printf("Enter the number of characters to be deleted\n");
  scanf("%d", &n);
  l = strlen(str);
  for(i = pos + n; i < l; i++)
  {
    str[i - n] = str[i];
  }
  str[i - n] = '\0';
  printf("The string is %s", str);
   return 0;
}
```

Output:


C:\Users\windows\Desktop\33.exe
Enter the string
hello
Enter the position where the characters are to be deleted
3
Enter the number of characters to be deleted
2
The string is hel
Process returned 0 (0x0)   execution time : 13.873 s
Press any key to continue.

16. Write a C program to determine if the given string is a palindrome or not (Spelled same in both directions with or without a meaning like madam, civic, noon, abcba, etc.)

Program:

```
#include <stdio.h>
#include <string.h>
int main() {
   char str[20];
   int i,j;
   int flag = 0;

   printf("Enter a string: ");
   scanf("%s", str);
    // Calculate the string length, -1 for '\0'
   j=strlen(str)-1;
   // Compare characters from the start and end of the string
   // and stop if a mismatch is found or the middle of the string is reached.
   for(i=0;i<=j;i++,j--)
   {
      if(str[i]!=str[j])
      {
         flag=1;
         break;
      }
   }
    if (flag) {
      printf("%s is not a palindrome\n", str);
   } else {
      printf("%s is a palindrome\n", str);
   }
}
```

Output:

```
Enter a string: madam
madam is a palindrome
```

17. Write a C program that displays the position of a character ch in the string S or – 1 if S doesn‘t contain ch.

Program:

```
#include <stdio.h>
#include <string.h>
int main() {
   char str[100], ch;
   char *position;
   // Input string and character
   printf("Enter the string: ");
   scanf("%[^\n]%*c", str);
   printf("Enter the character to find: ");
   scanf("%c", &ch);
   // Use strchr to find the first occurrence of ch in string S
   position = strchr(str, ch);
   // Check if character is found
   if (position != NULL) {
      // Output the index of the character
      printf("Position of '%c' in string: %d\n", ch, position - str);
   } else {
      printf("-1\n");  // If the character is not found
   }
   return 0;
}
```

Output:

```
Enter the string: welcome
Enter the character to find: c
Position of 'c' in string: 3
```

18. Write a C program to count the lines, words and characters in a given text.

Program:

```
#include <stdio.h>
int main() {
    // declare variables
   char str[200];
   int line, word, ch;

   // initialize count variables with zero
   line = word =1; ch = 0;

   // read multiline string
   printf("Enter string terminated with ~ :\n");
   scanf("%[^~]", str);

   for(int i=0; str[i]!='\0'; i++)
   {
      // if it is new line then
      // one line and one word completed
      if(str[i]=='\n')
      {
         line++;
         word++;
      }
      else  if(str[i]==' '||str[i]=='\t')
      {
         word++;
      }
      else
      {
         ch++;
      }
   }
   // display count values
   printf("\nCharacter counts = %d\n", ch);
   printf("Word counts = %d\n", word);
   printf("Line counts = %d\n", line);
   return 0;
}
```

Output

Enter string terminated with ~ :
hello world ~
Character counts = 10
Word counts = 3
Line counts = 1

Sorting and Searching:

19. Write a C program that uses non recursive function to search for a Key value in a given list of integers using linear search method.

Program:

```
#include<stdio.h>
int main()
{
 int i, a[20], n, key, flag = 0;
 printf("Enter the size of an array \n");
 scanf("%d", &n);
 printf("enter the array elements");
 for(i = 0; i < n; i++)
 {
  scanf("%d", &a[i]);
 }
  printf("enter the key elements");
 scanf("%d", &key);
 for(i = 0; i < n; i++)
 {
  if(a[i] == key)
  {
   flag = 1;
   break;
  }
 }
 if(flag == 1)
  printf("the key elements is found at location %d",i);
 else
  printf("the key elements is not found in the array");
 return 0;
}
```

Output:

```
Enter the size of an array
3
enter the array elements
41
12
14
enter the key elements
12
the key elements is found at location 1
```

20. Write a C program that uses non recursive function to search for a Key value in a given sorted list of integers using binary search method.

Program:

```
#include<stdio.h>
int main()
{
 int a[20], i, n, key, low, high, mid;
printf("enter the size of an array");
 scanf("%d",&n);
printf("enter the elements in ascending order");
 for(i = 0; i < n; i++)
 {
    scanf("%d",&a[i]);
 }
 printf("enter the key elements \n");
scanf("%d", &key);
 low = 0;
 high = n - 1;
 while(high >= low)
 {
  mid = (low + high) / 2;
  if(key == a[mid])
   break;
  else
  {
   if(key > a[mid])
    low = mid + 1;
   else
    high = mid - 1;
  }
 }
 if(key == a[mid])
 printf("The key element is found at location %d", mid+1);
 else
  printf("The Key Element is not found");
return 0;
}
```

Output:

```
enter the size of an array
5
enter the elements in ascending order
10
20
30
40
50
enter the key elements
30
The key element is found at location 3
```

21. Write a C program that implements the Bubble sort method to sort a given list of f. integers in ascending order

Program:

```
#include <stdio.h>
void swap(int* arr, int i, int j) {
    int temp = arr[i];
    arr[i] = arr[j];
    arr[j] = temp;
}
void bubbleSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        // Last i elements are already in place, so the loop
        // will only num n - i - 1 times
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1])
                swap(arr, j, j + 1);
        }
    }
}

int main() {
    int arr[] = { 6, 0, 3, 5 };
    int n = sizeof(arr) / sizeof(arr[0]);
    // Calling bubble sort on array arr
    bubbleSort(arr, n);
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```

Output:

0 3 5 6

22. Write a C program that sorts the given array of integers using selection sort in descending order.

Program:

```
#include <stdio.h>
// Utility function to swap two integer values
void swap(int *xp, int *yp) {
   int temp = *xp;
   *xp = *yp;
   *yp = temp;
}
// Implementation of selectionSortDesc to sort in descending order
void selectionSortDesc(int arr[], int n) {
   int i, j, max_idx;

   // One by one move the boundary of the unsorted subarray
   for (i = 0; i < n-1; i++) {
      // Find the maximum element in the unsorted array
      max_idx = i;
      for (j = i+1; j < n; j++) {
         if (arr[j] > arr[max_idx]) {
            max_idx = j;
         } }
      // Swap the found maximum element with the first element
      swap(&arr[max_idx], &arr[i]);
   }}
// Function to print the array
void printArray(int arr[], int size) {
   int i;
   for (i = 0; i < size; i++) {
      printf("%d ", arr[i]);
   }
   printf("\n");
}
// Main function to test the selectionSortDesc function
int main() {
   int arr[] = {64, 25, 12, 22, 11};
   int n = sizeof(arr)/sizeof(arr[0]);
   selectionSortDesc(arr, n);
   printf("Sorted array in descending order: \n");
   printArray(arr, n);
   return 0;
}
```

Output:

```
Sorted array in descending order:
64 25 22 12 11
```

23. Write a C program that sorts the given array of integers using insertion sort in ascending order.

Program:

```
#include <math.h>
#include <stdio.h>
void insertionSort(int arr[], int N) {

    // Starting from the second element
    for (int i = 1; i < N; i++) {
        int key = arr[i];
        int j = i - 1;
        // Move elements of arr[0..i-1], that are
          // greater than key, to one position to
          // the right of their current position
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j = j - 1;
        }
        // Move the key to its correct position
        arr[j + 1] = key;
    }
}
int main() {
    int arr[] = { 12, 11, 13, 5, 6 };
    int N = sizeof(arr) / sizeof(arr[0]);
    printf("Unsorted array: ");
    for (int i = 0; i < N; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
    // Calling insertion sort on array arr
    insertionSort(arr, N);
    printf("Sorted array: ");
    for (int i = 0; i < N; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
    return 0;
}
```

Output:

```
Unsorted array: 12 11 13 5 6
Sorted array: 5 6 11 12 13
```

24. Write a C program that sorts a given array of names.

Program:

```
#include <stdio.h>

#include <stdlib.h>
#include <string.h>
// Defining comparator function as per the requirement
static int myCompare(const void* a, const void* b)
{
    // setting up rules for comparison
    return strcmp(*(const char**)a, *(const char**)b);
}
// Function to sort the array
void sort(const char* arr[], int n)
{
    // calling qsort function to sort the array
    // with the help of Comparator
    qsort(arr, n, sizeof(const char*), myCompare);
}
int main()
{
    // Get the array of names to be sorted
    const char* arr[]
        = { "welcome", "hello", "greetings" };
    int n = sizeof(arr) / sizeof(arr[0]);
    int i;
    // Print the given names
    printf("Given array is\n");
    for (i = 0; i < n; i++)
        printf("%d: %s \n", i, arr[i]);
    // Sort the given names
    sort(arr, n);
    // Print the sorted names
    printf("\nSorted array is\n");
    for (i = 0; i < n; i++)
        printf("%d: %s \n", i, arr[i]);
    return 0;
}
```

Output:

Given array is
0: welcome
1: hello
2: greetings

Sorted array is
0: greetings
1: hello
2: welcome